Vision-Based Illegal Waste Detection and Reporting Using Geotagged Smartphone Images and YOLOv11.

Project Proposal



Supervisor:

DR S.S DUBE

Student:

PRESLEY NYAMA

N02539164N

The Department of Computer Science at the National University of Science and Technology in partial fulfilment of the requirements for the degree of

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1. Introduction

Illegal waste dumping has become a major environmental and public health challenge in Zimbabwe, particularly in rapidly growing urban centres such as Harare and Bulawayo. Increasing urbanisation, population growth, and limited municipal resources have overwhelmed existing waste management systems, leading to the rise of undesignated dumping sites in open spaces, along roadsides, and near water bodies. These sites degrade the urban environment, pollute soil and water sources, block drainage systems, and increase the risk of flooding during the rainy season (Fotovvatikhah et al., 2025).

The consequences of illegal dumping extend beyond environmental damage to serious public health and socio-economic impacts. Accumulated waste creates breeding grounds for disease vectors such as flies, rodents, and mosquitoes, contributing to outbreaks of illnesses including cholera, typhoid, and malaria. The burning of waste releases harmful pollutants that affect air quality and respiratory health. Furthermore, unmanaged dumping undermines sustainable urban development, reduces the aesthetic and economic value of neighbourhoods, and places additional strain on already stretched municipal services (Nwokediegwu et al., 2024).

Conventional monitoring approaches, which rely on manual inspections and public reporting, are often slow, reactive, and limited in coverage. Recent advances in Artificial Intelligence, particularly in computer vision and deep learning, offer an opportunity to improve environmental monitoring through automated image analysis (Fraternali, Morandini and González, 2024). This project proposes a vision-based approach that uses ordinary geotagged smartphone images, rather than specialised aerial equipment, combined with the YOLOv11 object detection model to detect and classify illegal waste dumping in real time. By pairing widely available mobile devices with geospatial technologies, the study aims to support data-driven municipal decision-making and contribute to a smarter, more affordable, and more sustainable model of urban waste management in Zimbabwe.

2. Background and Motivation

Zimbabwe's municipalities struggle with inefficient waste monitoring systems, which rely heavily on manual inspections and public reporting. These traditional approaches are reactive, slow, and resource-intensive, leaving many illegal dumping incidents unaddressed. At the same time, deep learning advancements, particularly Convolutional Neural Networks (CNNs) and the YOLOv11 detector, have proven highly effective in object detection and classification tasks, including waste recognition (Khanam and Hussain, 2024; Rajeev et al., 2025). Because smartphone ownership is now widespread across Zimbabwean cities and every modern handset already records GPS coordinates and a timestamp with each photograph, there is an opportunity to build a detection pipeline that needs no dedicated capture hardware at all. The motivation for this project therefore lies in combining an accessible image source, the smartphone camera, with state-of-the-art image recognition to meet Zimbabwe's pressing need for better urban waste management, without adding the cost, regulatory burden, or logistical overhead that specialised aerial equipment would otherwise require.

3. Problem Statement

Illegal waste dumping has become a serious challenge in Zimbabwe, especially in cities like Harare and Bulawayo. Piles of unregulated garbage are frequently discarded in undesignated areas, creating unsightly environments and blocking drainage systems. This practice contributes to flooding during the rainy season, spreads foul odours, and provides breeding grounds for harmful pests (Sesay and Fang, 2025). The resulting health hazards increase the risk of outbreaks of diseases like cholera, typhoid, and malaria, further straining Zimbabwe's healthcare system. Moreover, uncontrolled waste dumping pollutes soil and water, accelerates land degradation, and undermines environmental management efforts, posing long-term risks to both human well-being and ecological sustainability. Existing monitoring approaches remain manual and reactive, and proposals that depend on drones or satellite tasking introduce cost and regulatory barriers that are difficult to sustain in a resource-constrained research setting, which motivates a lower-cost, smartphone-based alternative.

4. Aim and Objectives.

4.1. Aim:

To design and implement an object recognition system that can detect and classify illegal waste dumping in Zimbabwe's urban environments using geotagged smartphone images, Convolutional Neural Networks, and YOLOv11.

4.2. Objectives:

The system is supposed to:

- Detect illegal waste dumping sites from smartphone photographs.
- Detect and classify waste by type.
- Determine and report the location of each detection to authorities using embedded GPS metadata.

5. Systematic Literature Review.

Illegal waste dumping and the use of deep learning for environmental monitoring have been increasingly studied in recent years. Research on object detection algorithms such as YOLO (You Only Look Once) and Convolutional Neural Networks (CNNs) highlights their applicability in real-world scenarios, including waste detection from ground-level, aerial, and satellite imagery.

Object Detection Models and Waste Recognition

Recent advancements in object detection have demonstrated significant improvements in speed and accuracy. Khanam and Hussain (2024) detail the architectural refinements introduced in YOLOv11, which improve small-object detection and inference efficiency compared with earlier YOLO releases, an important factor when identifying dispersed waste items in cluttered urban scenes. Building on this, Nasien et al. (2025) applied YOLOv11 directly to waste classification and reported strong accuracy across common recyclable categories, while Rajeev et al. (2025) customised YOLO-based architectures for e-waste classification with promising results. Xia et al. (2024) further proposed YOLO-MTG, a lightweight YOLO variant tuned specifically for multi-target garbage detection, demonstrating that recent YOLO iterations are well suited to the varied, low-resolution conditions typical of citizen-submitted photographs rather than only high-resolution aerial captures.

Deep Convolutional Neural Networks for Waste Classification

Convolutional Neural Networks remain foundational for feature extraction and classification. Abo-Zahhad et al. (2025) combined two-stage cascaded YOLOv5 and YOLOv8 classifiers to distinguish trash bags, overflowing bins, and roadside litter in real time, achieving high precision even on modest hardware. Sesay and Fang (2025) reviewed innovations in municipal solid waste classification and noted that CNN-based pipelines trained on augmented, locally sourced datasets consistently outperform generic pretrained models when applied to context-specific waste streams, reinforcing the value of building a Zimbabwe-specific image dataset rather than relying solely on datasets collected elsewhere.

Smartphone and Citizen-Sensing Approaches to Waste Detection

A growing body of work shows that smartphone-captured imagery is a practical and sufficient input for waste detection pipelines, removing the need for dedicated aerial hardware. Moral et al. (2022) demonstrated a computer-vision pipeline for waste-container localisation that geo-positions detections directly onto city maps using image metadata, an approach directly transferable to individually captured smartphone photographs. More recently, mobile-based implementations have paired YOLOv11n with on-device inference frameworks such as TensorFlow Lite to classify household waste type and condition directly from phone cameras, reporting results comparable to server-side models while requiring no specialised capture equipment. Alharbi et al. (2025) similarly used deep learning on street-level imagery to detect public littering behaviour in real time, illustrating that ground-level, citizen-generated images carry sufficient visual detail for reliable detection. Taken together, this literature indicates that the detection accuracy of a well-trained YOLOv11 model depends far more on dataset quality and annotation than on the specific camera platform used to capture the images.

Aerial and Satellite-Based Approaches: Prior Work and Their Limitations for This Context

Aerial and satellite imagery have also been explored for illegal dumping detection. Padubidri, Kamilaris and Karatsiolis (2022) used high-resolution aerial photography with deep learning to detect illegal dumping sites, while Torres and Fraternali (2021) applied scene classification to aerial images to identify landfills, and Youme et al. (2021) paired UAV imagery with deep learning models for the same purpose. Didelija et al. (2022) additionally showed that detection accuracy on satellite imagery is highly sensitive to segmentation-scale parameters, requiring careful site-specific tuning. Fraternali, Morandini and González (2024), in a broader survey of remote-sensing-based waste detection, observed that aerial and satellite approaches consistently face three recurring constraints: the cost of acquiring stable, high-resolution, georeferenced imagery; regulatory approval requirements for UAV flights over populated areas; and reduced visibility of waste that is sited behind buildings, under tree cover, or along narrow alleys, precisely the locations where illegal dumping in Harare and Bulawayo is most often observed. These constraints motivate the use of ground-level, geotagged smartphone images as the primary data source for this project, since they avoid airspace permissions entirely and often capture more diagnostic detail of waste at eye level than a downward-facing aerial sensor would.

Geospatial Integration and Smart Urban Management

Integrating object detection with Geographic Information Systems (GIS) enables spatial analysis and actionable decision support. Moral et al. (2022) combined detection outputs with automatic geo-positioning to visualise container and waste hotspots on city maps, while Nwokediegwu et al. (2024) compared AI-driven waste management innovations across different regions and highlighted geospatial dashboards as a recurring enabler of faster

municipal response. This integration underpins the smart-city agenda, where real-time environmental monitoring, built from data that citizens and municipal staff already generate with their own phones, contributes to sustainable urban management.

Research Gaps

Despite this progress, limited studies focus specifically on Zimbabwe, and most existing smartphone-based or citizen-sensing waste detection studies were conducted in Asia, the Gulf region, or East Africa (Fotovvatikhah et al., 2025; Alharbi et al., 2025). There is a need for region-specific datasets, low-cost implementations suited to resource-constrained municipalities, and evaluations of YOLOv11 performance under Zimbabwe's specific environmental, lighting, and infrastructural conditions. This project addresses these gaps by applying a smartphone-based detection pipeline to Zimbabwean urban environments, deliberately avoiding the cost and regulatory overhead associated with aerial data collection.

6. Methodology

6.1. Research Method:

This project will adopt Design Science Research (DSR) as the research methodology because it is best suited for developing and evaluating an artifact that solves a real-world problem. It is a distinct research paradigm that can employ any of these methodological approaches depending on the nature of the artifact being developed and how it is evaluated. DSR provides a structured approach to problem identification, artifact design, demonstration, and rigorous evaluation, making it ideal for building and testing an AI-based waste detection system in Zimbabwe's context.

6.2. Software Method:

For the software methodology, the project will use an Agile approach because it allows iterative development, flexibility, and continuous improvement. Given that deep learning projects often require frequent adjustments in datasets, model parameters, and deployment strategies, Agile enables the system to be built in small sprints, tested quickly, and refined based on performance results.

7. Tools and Technologies to be Used

The project will utilise a combination of freely available hardware and software tools to develop and deploy the proposed waste detection system. Data will be collected using ordinary Android and iOS smartphones, which are already widely owned across Harare and Bulawayo; each photograph captured will carry embedded GPS coordinates and a timestamp in its EXIF metadata, removing the need for any additional positioning hardware. YOLOv11, a state-of-the-art real-time object detection algorithm (Khanam and Hussain, 2024), will be used to identify and locate waste piles in the collected images with high speed and accuracy. A Convolutional Neural Network will then classify the detected waste into categories such as plastic, organic, metal, tyres, construction waste, and mixed waste to support targeted municipal responses. Images will be annotated using free-tier tools such as LabelImg, CVAT, or Roboflow. The system will be developed in Python, which provides powerful machine learning libraries including PyTorch, OpenCV, and NumPy for model implementation and image processing, with model training carried out on Google Colab's free GPU tier to avoid the cost of dedicated hardware. A database management system, such as SQLite or

PostgreSQL, will store geospatial data, detection results, image metadata, and system logs. Geographic Information System tools such as QGIS and the Python Folium library will support spatial analysis, mapping of dumping locations, and the creation of interactive visualisations for authorities. A lightweight backend built with Flask or FastAPI will connect detection results to the mapping layer, and Git and GitHub will be used for version control, enabling efficient code management, collaboration, and tracking of project development throughout the lifecycle.

8. Scope of the Study

This study focuses on the design, development, and evaluation of an intelligent computer vision system for detecting and classifying illegal waste dumping in selected urban areas of Zimbabwe, particularly Harare and Bulawayo. The project is limited to the use of geotagged smartphone photographs, which will be analysed using YOLOv11 for object detection and Convolutional Neural Networks for waste classification. The study covers dataset collection, model training, testing, and performance evaluation using annotated images of common waste types such as plastic, organic, metal, tyres, construction waste, and mixed waste. It also includes the integration of geospatial tools for mapping detected dumping sites and generating reports for municipal authorities. However, the study does not cover physical waste removal, policy enforcement, or the development of full-scale municipal infrastructure, as its primary focus is on the technological detection and decision-support component.

9. Feasibility Study

Technical Feasibility: Advances in computer vision and the availability of open-source deep learning frameworks (e.g. PyTorch, TensorFlow) make the development of a smartphone-based waste detection system technically viable. Modern smartphones, whether Android or iOS, already produce sufficiently high-resolution images with embedded GPS and timestamp metadata, and free-tier GPU resources on Google Colab support model training without local hardware investment.

Operational Feasibility: Municipal stakeholders are increasingly open to technology-driven solutions to environmental challenges. Because data collection relies on ordinary smartphones rather than specialised equipment, no pilot training, flight scheduling, or equipment maintenance is required, and images can be gathered by students, volunteers, or eventually members of the public through a simple reporting interface. Integration with existing municipal workflows is feasible with moderate adaptation.

Economic Feasibility: Because the project uses devices the student and volunteers already own, together with free-tier annotation, training, and mapping tools, the direct cost of building the dataset and training the model is minimal. The long-term benefits, reduced manual monitoring costs, faster response times, and improved environmental outcomes, further justify the modest investment required, and partnerships with environmental NGOs or municipal authorities could support future scaling.

Legal and Ethical Feasibility: Because data collection uses handheld smartphones in public spaces rather than UAVs, the project falls outside the scope of Zimbabwe Civil Aviation Authority (ZCAA) drone regulations, considerably simplifying the legal and administrative burden of data collection. Ethical considerations remain important: images will be processed with a focus on waste and public spaces rather than identifiable individuals, faces or number

plates incidentally captured will be blurred or excluded during annotation, and data storage will be kept secure.

10. Area of Interest.

This research falls within the interdisciplinary area of Artificial Intelligence, with a particular focus on computer vision and deep learning for environmental monitoring. It explores how Convolutional Neural Networks and real-time object detection models such as YOLOv11 can be applied to analyse visual data captured on ordinary smartphones and automatically identify patterns related to illegal waste dumping. The study also intersects with Geographic Information Systems (GIS) and smart city technologies, as it combines image analysis with spatial metadata to support informed decision-making. More broadly, the research contributes to the growing field of Environmental Informatics, where advanced computing techniques are used to address environmental and urban management challenges through intelligent, data-driven systems.

11. Area of Application

The application area of this study lies in urban environmental management and municipal service delivery. The proposed system can be used by local authorities, or eventually by citizens through a simple reporting application, to monitor illegal waste dumping sites more efficiently through smartphone-based image capture and automated analysis. By providing accurate detection, classification of waste types, and geospatial mapping of dumping locations, the system supports faster response, improved waste management planning, and better allocation of municipal resources. Additionally, the solution has practical relevance in public health protection, pollution control, and smart city development, where digital technologies are increasingly used to improve urban living conditions and environmental sustainability.

12. Project Significance

This project is significant as it addresses a pressing environmental and public health problem in Zimbabwe using advanced artificial intelligence techniques that are affordable and readily deployable. By automating waste detection from devices that citizens and municipal staff already own, the system has the potential to support municipalities in monitoring and responding to illegal dumping more effectively, thereby reducing disease risks, preventing flooding, and promoting cleaner urban environments. Because the approach removes the cost and regulatory burden of aerial data collection, it is particularly well suited to resource-constrained settings, and the project contributes to Zimbabwe's journey toward smart city initiatives by demonstrating how AI can be harnessed for sustainable urban management without specialised equipment.

13. Expected Outcome

The expected outcome of this study is the successful development of an intelligent computer vision system capable of accurately detecting and classifying illegal waste dumping in urban areas of Zimbabwe. The system is anticipated to demonstrate strong performance in identifying waste piles from geotagged smartphone images using YOLOv11, even in complex and

cluttered urban environments. Detection accuracy is expected to exceed 80%, with reliable localisation of dumping sites through GPS metadata already embedded in each captured image and displayed through GIS mapping.

In addition, the Convolutional Neural Network component of the system is expected to effectively classify different waste categories such as plastic, organic, metal, tyres, construction waste, and mixed waste with a high level of precision. This classification capability will provide valuable information for municipal authorities to prioritise clean-up efforts and allocate resources more efficiently.

The integration of detection results with Geographic Information Systems is expected to produce interactive spatial maps highlighting illegal dumping hotspots in cities such as Harare and Bulawayo. These visual outputs will support data-driven decision-making, enabling quicker response times and improved environmental monitoring.

From a research perspective, the project is also expected to contribute a locally relevant, low-cost, and reproducible annotated dataset of illegal waste images collected entirely from smartphones, which can support future studies in Zimbabwe and similar developing-country contexts. Furthermore, performance evaluation of YOLOv11 on smartphone-sourced imagery in this environment will add to academic knowledge on applying deep learning to environmental challenges in resource-constrained settings.

Overall, the system is expected to demonstrate that artificial intelligence, paired with devices people already carry, can provide a practical, scalable, and cost-effective tool for improving urban waste management, supporting public health, and advancing smart city initiatives in Zimbabwe.

14. Budget

The estimated budget for this project covers the modest operational costs required to develop and deploy the proposed waste detection system, since it deliberately avoids specialised capture hardware. Data collection will use the student's own Android and iOS smartphones, together with volunteers' devices where needed, at no equipment cost. Model training and experimentation will be carried out primarily on Google Colab's free GPU tier; a contingency of approximately USD 50 to USD 100 is budgeted for optional Colab Pro credits or cloud storage should larger datasets or longer training runs be required.

Annotation will use the free tiers of Roboflow and CVAT, which are sufficient for a dataset of several thousand images, so no annotation licensing cost is anticipated. Mapping and spatial analysis will rely on the open-source QGIS application and the Python Folium library, both free of charge. Fieldwork remains a necessary component of the project; transport and data-bundle costs for travelling around Harare and Bulawayo to photograph dumping sites, clean areas, bins, and other reference locations are estimated at USD 60 to USD 100 over the project duration. A small contingency of USD 30 to USD 50 is also set aside for incidental costs such as printing, internet connectivity, or minor phone accessories.

Overall, the total projected budget is significantly lower than a drone-based equivalent and falls comfortably within the means of a self-funded Master's student, reflecting the cost-efficiency that a smartphone-based data collection strategy provides while still delivering a functional and reliable intelligent waste detection system.

15. Time Frame

The project is expected to run over approximately six months. The first month will cover literature consolidation and finalisation of the research design. The second and third months will focus on smartphone-based data collection across Harare and Bulawayo and annotation of the resulting image dataset. The fourth month will be dedicated to training and tuning the YOLOv11 detection model and the CNN classification component. The fifth month will focus on integrating detection outputs with GIS mapping and building the reporting interface, followed by system testing and evaluation. The final month will be used for performance analysis, write-up, and dissertation submission.

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